HOUSE PRICE PREDICTION USING MACHINE LEARNING

Project ID: PRCP-1020

Dataset: Ames, Iowa Housing Dataset

Project Type: Supervised Machine Learning — Regression

1. Executive Summary

This project focuses on predicting residential house sale prices using machine learning techniques and the Ames, Iowa housing dataset. The dataset contains 1,460 records, 80 explanatory variables, and the target variable `SalePrice`.

The project includes data auditing, exploratory data analysis (EDA), missing-value treatment, categorical encoding, feature engineering, model training, hyperparameter tuning, cross-validation, and prediction error analysis. Random Forest and Gradient Boosting regression models were evaluated.

The baseline Gradient Boosting model achieved a Mean Absolute Error (MAE) of approximately $16,570.52, a Root Mean Squared Error (RMSE) of approximately $26,577.07, and an R^2 score of 0.908 on the selected test set. It was the leading candidate among the evaluated configurations based on the recorded test results.

2. Business Understanding and Objectives

House prices depend on several characteristics, including living area, overall quality, garage capacity, basement area, and location. Estimating prices consistently can support preliminary property pricing and analysis.

The main objective of this project is to develop a machine learning regression model that predicts house sale prices from available property characteristics.

The objectives are:

- Understand and audit the housing dataset.
- Explore relationships between property features and sale prices.
- Handle missing values and categorical variables.
- Train and compare Random Forest and Gradient Boosting models.
- Evaluate models using MAE, RMSE, and R^2.
- Test feature engineering and hyperparameter tuning.
- Analyze prediction errors and identify limitations.
- Save the leading model for future use.

3. Dataset Description

The project uses the Ames, Iowa housing dataset.

Dataset property	Description
Number of records	1,460
Original columns	81
Explanatory columns	80 before removing the identifier
Target variable	`SalePrice`
Duplicate rows	0
Missing target values	0

The `SalePrice` variable represents the sale price of each property and is the value the model attempts to predict.

4. Data Quality Assessment

The initial audit found no duplicate rows and no missing values in the target variable. However, several explanatory columns contained missing values.

Some of the columns with the highest missing-value counts were:

- `PoolQC`: 1,453 missing values (99.52%).
- `MiscFeature`: 1,406 missing values (96.30%).
- `Alley`: 1,369 missing values (93.77%).
- `Fence`: 1,179 missing values (80.75%).
- `MasVnrType`: 872 missing values (59.73%).

Missing values may sometimes indicate that a property does not have a particular feature. Therefore, missingness should be interpreted carefully rather than automatically treated as an error.

5. Exploratory Data Analysis

The target variable `SalePrice` had the following descriptive statistics:

Statistic	Value
Mean	$180,921.20
Median	$163,000
Minimum	$34,900
Maximum	$755,000
First quartile	$129,975
Third quartile	$214,000
Skewness	1.883

The mean is higher than the median, and the skewness is positive. This indicates a right-skewed distribution, with some properties selling at substantially higher prices than most properties.

The strongest recorded numerical correlations with `SalePrice` included:

Feature	Correlation
`OverallQual`	0.791
`GrLivArea`	0.709
`GarageCars`	0.640
`GarageArea`	0.623
`TotalBsmtSF`	0.614
`1stFlrSF`	0.606
`FullBath`	0.561
`TotRmsAbvGrd`	0.534
`YearBuilt`	0.523
`YearRemodAdd`	0.507

These correlations suggest that overall quality, living area, garage capacity, basement area, and floor area are associated with sale prices. Correlation does not prove causation, and these numerical results do not describe every relationship in the dataset.

6. Data Preprocessing

The following preprocessing steps were applied:

1. The target variable `SalePrice` was separated from the input features.
2. The identifier column `Id` was removed from the predictors.
3. The data was divided into training and test sets using an 80/20 split with `random_state=42`.
4. Numerical missing values were handled using median imputation.
5. Categorical missing values were handled using most-frequent-value imputation.
6. Categorical variables were transformed using one-hot encoding.
7. Preprocessing and model training were combined in scikit-learn pipelines.

The resulting dataset contained 79 predictor features. The training set contained 1,168 records, while the test set contained 292 records.

Using pipelines helps ensure that preprocessing transformations are fitted within the training workflow rather than manually applied to the entire dataset before splitting.

7. Model Development

Two regression algorithms were selected.

Random Forest Regressor: This algorithm combines predictions from multiple decision trees. It can model nonlinear relationships and interactions among property features.

Gradient Boosting Regressor: This algorithm builds a sequence of decision trees, with later trees helping reduce errors made by earlier ones.

Both models were trained with baseline parameters. Hyperparameter tuning was also performed using `RandomizedSearchCV`, three-fold cross-validation, and negative RMSE as the scoring metric.

8. Baseline and Tuned Model Evaluation

The recorded test-set results were:

Model	MAE	RMSE	R^2
Baseline Gradient Boosting	$16,570.52	$26,577.07	0.908
Baseline Random Forest	$17,386.11	$28,495.23	0.894
Tuned Gradient Boosting	$16,631.39	$27,400.00	0.902
Tuned Random Forest	$17,365.53	$28,881.74	0.891

MAE represents the average absolute difference between actual and predicted prices. RMSE penalizes larger errors more strongly. R^2 measures how much of the target's variation is explained by the model relative to a mean-based baseline.

Among these four recorded configurations, baseline Gradient Boosting had the lowest MAE and RMSE and the highest R^2. The tuned versions did not improve test-set metrics over their respective baseline models.

Therefore, baseline Gradient Boosting was retained as the leading candidate based on the recorded comparison.

9. Cross-Validation Analysis

Five-fold cross-validation was used to compare the baseline Random Forest and Gradient Boosting models.

Model	Mean MAE	Mean RMSE	Mean R^2	RMSE standard deviation
Random Forest	$18,162.79	$30,168.27	0.845	$4,933.87
Gradient Boosting	$16,140.98	$27,900.41	0.865	$6,321.98

Gradient Boosting performed better on the recorded mean MAE, mean RMSE, and mean R^2 values. However, its RMSE standard deviation was higher, indicating greater variation across folds.

Cross-validation provides a broader view of model performance than a single train/test split, although the results still depend on the data and evaluation procedure.

10. Feature Engineering

Three additional features were created to test whether combining related property measurements could improve predictions.

- `TotalSF`: Sum of basement, first-floor, and second-floor area.
- `TotalBathrooms`: Sum of full bathrooms and half-weighted half bathrooms, including basement bathrooms.
- `TotalPorchSF`: Sum of selected porch and deck area measurements.

The recorded results were:

Feature set	MAE	RMSE	R^2
Original features	$16,570.52	$26,577.07	0.908
Engineered features	$16,190.89	$26,669.75	0.907

The engineered model reduced MAE slightly, but RMSE increased slightly and R^2 decreased slightly. The outcome was mixed, so feature engineering was not considered an improvement across all metrics.

11. Feature Importance

Feature importance was extracted from the fitted baseline Gradient Boosting pipeline after preprocessing.

The resulting importance table is saved as `outputs/feature_importance.csv`.

Feature importance can help identify which transformed inputs the model relied on most when making predictions. These values describe the model's predictive behavior; they do not establish that a feature causes a change in house prices.

The saved CSV should be consulted for the exact ranking and importance values before making detailed claims about the top model features.

12. Prediction Error Analysis

The baseline Gradient Boosting model's test-set error statistics were:

- MAE: approximately $16,570.52.
- RMSE: approximately $26,577.07.
- Mean signed residual (actual minus predicted): approximately +$336.40.
- Median absolute error: approximately $11,542.00.
- Largest absolute error: approximately $191,238.43.

The mean signed residual was close to zero, but this does not mean that individual predictions were consistently accurate. Some properties had much larger errors than the typical case.

The largest-error cases included both under-predictions and over-predictions. Such cases should be examined for unusual property characteristics, rare high prices, data-quality issues, or patterns that the model does not represent well. The residual values alone do not identify the cause of each error.

13. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. **Use predictions as preliminary estimates.** The model can support initial pricing analysis but should not replace professional valuation or local market knowledge.
2. **Review important property characteristics.** Overall quality, living area, garage features, and basement and floor area showed notable associations with sale prices.
3. **Investigate high-error properties.** Review unusual properties and large prediction errors before using estimates in important decisions.
4. **Validate on newer data.** Housing markets change over time, so the model should be evaluated on more recent and representative sales.
5. **Monitor performance.** If the model is used in a real application, track prediction errors and retrain only when supported by suitable data and evaluation.
6. **Document the model.** Record the model version, data source, preprocessing steps, limitations, and evaluation metrics.

14. Limitations

The project has several limitations:

- The dataset represents houses in Ames, Iowa, so results may not generalize to other locations or current market conditions.
- A single random train/test split may not represent every future property.
- The test set was used in repeated model comparisons. Consequently, it was not kept completely untouched, and the reported test metrics should be treated as exploratory rather than a fully independent final performance estimate.
- Missing-value treatment and encoding choices can affect results.
- Correlation and feature importance do not establish causation.
- The model has not been established as production-ready. External validation, monitoring, and a documented prediction workflow would be needed before real-world deployment.

15. Saved Model and Project Outputs

The leading candidate model was saved as a scikit-learn pipeline at:

`models/house_price_gradient_boosting.joblib`

The main notebook is:

`PRCP-1020_House_Price_Prediction.ipynb`

The dataset is stored at:

`Data/data.csv`

The `outputs` folder contains the following evaluation files:

- `model_comparison.csv`
- `cross_validation_comparison.csv`
- `feature_importance.csv`
- `prediction_error_analysis.csv`

- `top_error_analysis.csv`

These files support review of the model comparison, cross-validation, feature importance, and prediction errors.

16. Conclusion

This project completed an end-to-end machine learning workflow for house price prediction using the Ames, Iowa housing dataset. The work included data auditing, exploratory data analysis, preprocessing, baseline model comparison, hyperparameter tuning, cross-validation, feature engineering, error analysis, and model saving.

Among the evaluated configurations, baseline Gradient Boosting was the leading candidate. It achieved a recorded test RMSE of approximately $26,577.07 and an R^2 score of 0.908. Feature engineering produced mixed results, and hyperparameter tuning did not improve the recorded test metrics.

The project demonstrates how regression models can be developed and evaluated for a structured prediction task. Further validation on independent, current, and geographically appropriate data is required before practical deployment.

17. Final Submission Checklist

Before submission:

- Confirm that this report matches the assignment's required format.
- Add candidate name, institution, or other identifying project details if required by the official instructions.
- Confirm that all notebook cells run without errors and save the notebook.
- Verify the dataset, saved model, and five CSV output files.
- Include charts and captions if the assignment specifically requires them.
- Check the official assignment portal for required file formats, submission instructions, and deadline.
- Submit every deliverable explicitly requested by the assignment guidelines.

End of Report